

AutoAstro: Automating Data Analysis in Astronomy Using Large Language Models

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Abstract

With the explosive growth of astronomical data, the demand for simplified, flexible tools to process and analyze them has steadily increased. Recently, large language models (LLMs) have demonstrated remarkable reasoning and planning capabilities for language-related scientific tasks. They show great potential to automate data analysis and experimentation, thereby reducing human effort and enhancing research efficiency. In this paper, we introduce an LLM-empowered framework, AutoAstro, for automated data analysis in astronomy. Specifically, AutoAstro can instruct LLMs to select appropriate tools from prebuilt libraries and construct a unified data analysis workflow based on users' requirements described in natural language. We present AutoAstro's abilities to produce accurate visual and textual outputs aligned with user intent across multiple astronomical tasks on a curated benchmark. Ultimately, AutoAstro aims to serve as a tool to increase the productivity of astronomers, especially those without programming expertise, by reducing their data processing burden.

Keywords: astroinformatics, large language models, artificial intelligence for science,
automated workflow construction

1 Introduction

Recent decades have witnessed unprecedented growth of astronomical data due to the deployment of ground-based and space-based telescopes and the implementation of large-scale sky surveys, e.g., SDSS [1], LAMOST [2], *Gaia* [3], and LSST [4]. Such massive data, on the one hand, significantly drive astronomical discovery by facilitating the understanding of the universe using computational methods on observational datasets, and on the other hand, pose enormous challenges to analytical techniques, as rapidly generated TB-to-PB-scale data have overwhelmed existing systems and made it imperative to design novel, more efficient data analysis frameworks [5].

To meet the needs of data-intensive astronomical tasks, astroinformatics, an interdisciplinary field that integrates astronomy, data science, and artificial intelligence (AI) to extract, analyze, and interpret information from astronomical data, has emerged in recent years. The applications of astroinformatics span a wide range of problems, including spectral analysis [6], time-series forecasting [7], object identification [8], stellar parameter estimation [9], anomalous event detection [10], and morphological classification [11], significantly enhancing the efficiency of astronomical discovery. To address these diverse and complex tasks, intelligent computational methods, including machine learning (ML), deep learning (DL), statistical analysis, and visualization techniques, have been widely adopted. Existing studies have demonstrated the efficacy of machine learning models [12] for astronomical classification and regression based on numerical attributes, including star-galaxy separation, photometric redshift estimation, and exoplanet candidate classification. Meanwhile, deep learning methods [13] have achieved exceptional performance in astronomical image and time-series analysis tasks. Statistical methods [14] also play a fundamental role in astronomical correlation analysis, parameter estimation, and uncertainty quantification. These methods extract valuable information from large-scale data, enabling the construction of data-driven astronomical models and deepening the understanding of cosmic structure and evolution [15]. However, integrating them into a unified framework for adaptive model selection and combination based on data characteristics and scientific goals still poses substantial challenges. Additionally, the development and deployment of these methods in real-world astronomical applications often require specialized domain expertise and extensive experimentation. Although there have been pre-built toolchains, such as scikit-learn [16], Astropy [17], and PyTorch [18], they still impose high requirements on programming skills and pose usability barriers for non-programmers.

With the growing demand for scalable data analysis in astronomy, the field of astroinformatics has advanced rapidly towards more intelligent and automated data analysis workflows [19–21]. Recent advances in large language models (LLMs), e.g., ChatGPT [22], offer new opportunities for scientific workflow automation. Pretrained

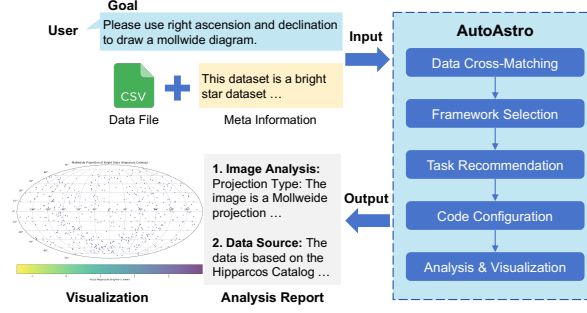


Fig. 1 Overview of AutoAstro. We show an exemplar astronomical task and the sequence of its execution steps using AutoAstro. Based on a user-specified task description, AutoAstro proceeds through data cross-matching, framework selection, task recommendation, code configuration, and analysis & visualization. The final output comprises an analysis report accompanied by a visual plot.

on hundreds of billions of samples from diverse sources, LLMs demonstrate remarkable performance in semantic understanding, logical reasoning, and executable code generation, unlocking levels of automation previously unattainable. In particular, LLM-based agent frameworks, such as ReAct [23] and TaskWeaver [24], are emerging as transformative tools that incorporate reasoning into decision-making loops and build workflows for complex tasks across scientific and engineering domains with their planning, execution, and iterative refinement capabilities. For example, commercial LLM services, such as GPT-4o [25], Gemini [26], and AutoGLM [27], offer user-friendly data analysis features. However, these services still have several restrictions, including a lack of support for secondary development, time-consuming online interactions, and insufficient integration with domain-specific tools. AstroML [28] provides a more flexible, customizable data analysis framework. However, the AstroML workflow must still be built manually by users with programming skills. Agent Poirot [29] and DS-Agent [30] provide end-to-end data analysis pipelines that offer user-friendly input and support for secondary development. In these frameworks, users only need to specify the goal of the data analysis and the data to be used, significantly reducing the time and effort required. However, they are only applicable for basic machine learning model training and cannot fulfill the requirements of astronomical data analysis.

Very recently, LLMs and LLM-based agent systems have been developed for astronomical research. First, LLMs have been applied to entity recognition and information extraction from astronomical literature [31–33]. Second, vision language models (VLMs) have been leveraged to assist in the captioning and interpretation of astronomical images [34–36]. Third, LLM-based agents have been developed to integrate with telescopes for automated instrument control and real-time data processing [20, 21]. They have substantially reduced the burden of research workflows, thereby allowing astronomers to focus on core scientific questions rather than tedious tasks. However, they still focus on a single step of the astronomical research (e.g., literature review, image analysis, or instrument control) and cannot yet implement an integrated end-to-end data analysis pipeline.

To address these gaps, we present an LLM-based agent framework, AutoAstro, to automate the scientific data analysis workflow in astronomy. AutoAstro provides a lightweight and flexible pipeline for astronomical data analysis with the aim of enhancing research efficiency. As illustrated in Fig. 1, AutoAstro enables automated analysis across diverse astronomical tasks by streamlining user task descriptions into three key components: data file, data description, and analysis goals. Then, AutoAstro’s agents autonomously formulate analysis tasks, generate executable code and tool calls, execute data processing, and produce visualizations and reports via a prompt pipeline optimized for astronomical tasks. In general, AutoAstro makes the following two key contributions. First, it proposes a novel LLM-based agent tailored to the astronomy domain that can automatically perform data analysis, visualization, and modeling tasks. Second, it implements a lightweight, end-to-end, and intelligent data analysis platform for astronomy.

To evaluate AutoAstro, we utilize the curated scientific evaluation benchmark NL2Astro, comprising 17 datasets and 48 natural-language task instructions. These tasks, spanning multiple domains and levels of complexity, are designed to present typical research problems on astronomical data analysis and have been manually checked for rigorous evaluation. Quantitative results demonstrate that AutoAstro achieves a success rate of approximately 91% across all tasks, which is only 9% lower than the perfect pass rate achieved by human experts and at least 3% higher than that of state-of-the-art LLM-based approaches. Furthermore, in terms of task performance scores, AutoAstro outperforms human experts by 6.69% and surpasses any other LLM-based approach by at least 55%. In terms of efficiency, AutoAstro reduces time consumption by approximately 90% compared to the average of 82 minutes per task by human experts. Moreover, we validate the effectiveness of AutoAstro through case studies across three distinct scenarios: validating the inverse-square law of brightness, constructing a machine learning model for spectral classification, and predicting stellar parameters. These cases further confirm that AutoAstro can autonomously build and execute complex, multi-tool workflows while maintaining the accuracy of the generated models.

2 Results

2.1 Overview of AutoAstro

AutoAstro is an LLM-based agent specifically designed for astronomical data analysis. The overall framework design is inspired by LightVA [37]. Its primary goal is to construct an end-to-end workflow by integrating data processing, modeling, analysis, and visualization into a unified platform. To achieve this, AutoAstro consolidates multiple functionally distinct code modules to facilitate coordinated and synergistic operation. As illustrated in Fig. 2, the tool sets in AutoAstro include those for data cross-matching, statistical analysis, machine-learning modeling, deep-learning modeling, and visual analytics, all orchestrated by an LLM.

The overall workflow consists of three sequential stages. In the first stage, user input is processed into three essential elements: the scientific objective, the dataset, and its associated metadata. AutoAstro then performs data cross-matching, selects

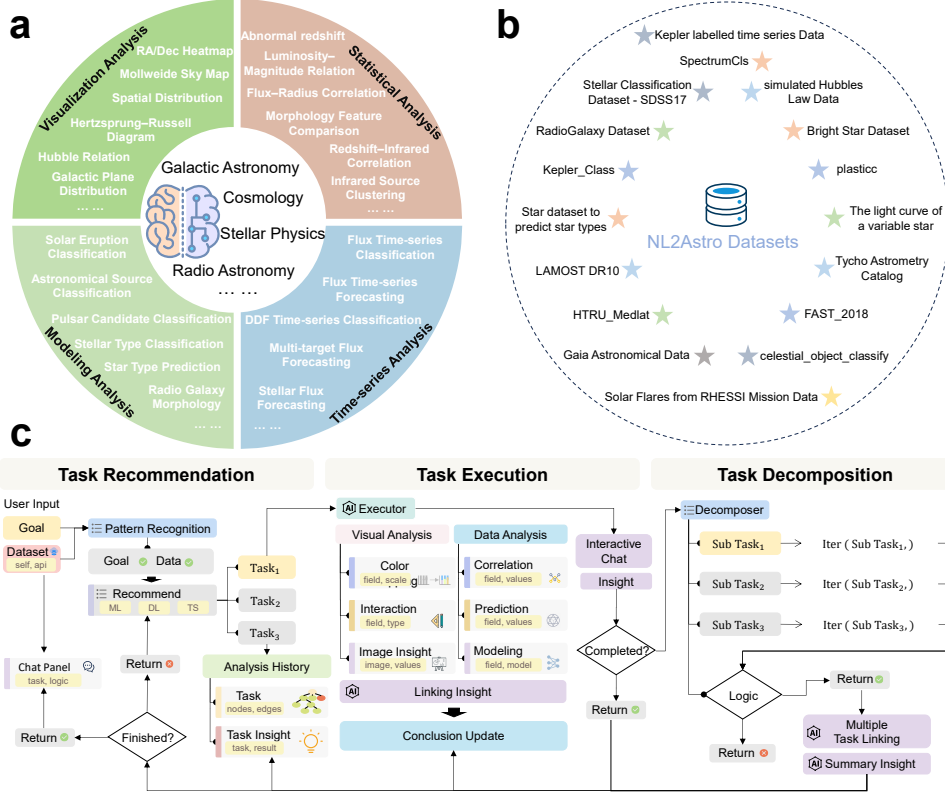


Fig. 2 Overall Framework of AutoAstro. Fig. a) illustrates the astronomical sub-domains covered by AutoAstro and the main analytical tasks. It encompasses diverse workflows, including the modeling and analysis for pulsar classification and flare identification, statistical investigations of redshift and spatial distributions, and visualization of sky regions and galaxy-cluster sources. In addition, it also supports time-domain analytical tasks related to Flux and Deep Drilling Fields (DDF), including light-curve modeling, temporal feature extraction, and the characterization of long-term observational variability. Fig. b) presents the datasets within NL2Astro, which include large-scale observational datasets derived from major astronomical surveys and facilities, such as LAMOST, FAST, and SDSS, as well as datasets for validation and methodological assessment, including simulated Hubble’s law data and the stellar dataset. Fig. c) illustrates the architecture of AutoAstro: Stage 1 interprets the user’s objectives while maintaining an analytical history to ensure traceability and reproducibility; Stage 2 performs in-depth task analysis and reasoning, synthesizes the analytical outputs, and dynamically updates intermediate conclusions as needed; Stage 3 evaluates the completeness of the current analytical process, and further decomposes complex objectives into finer-grained sub-tasks when required, enabling recursive planning and efficient execution.

an execution framework, and recommends feasible tasks to the user. In the second stage, AutoAstro invokes the appropriate analytical modules for the selected task, generates the code to be executed, and obtains the output. In the third stage, AutoAstro evaluates whether the output satisfies the initial user-specified goals. If not, it recursively decomposes the task into smaller subtasks, refines the analysis procedure to regenerate the code, and updates the output accordingly. Additionally, the workflow incorporates an error-detection and correction mechanism. Upon any failure, AutoAstro retries the above procedure until an upper limit is reached.

The AutoAstro framework is formally denoted as a function f that maps a user-specified input text T and an astronomical dataset D into the output $\hat{Y}$ for a task identified from T and performed on D , which can be images, plots, numeric estimations, labels, or other computational artifacts depending on the task type, and an analysis report R in natural language, where an LLM L serves as the core controller to coordinate the operation process, i.e.,

$$f(T, D | L) \rightarrow (\hat{Y}, R). \quad (1)$$

In general, AutoAstro integrates LLM-driven reasoning with modular computational modeling, enabling interpretable, scientifically grounded processing of astronomical data analysis tasks.

2.2 Performance Evaluation of AutoAstro

2.2.1 Evaluation setup

Dataset: To evaluate the performance of AutoAstro, we created NL2Astro, a dataset comprising 48 astronomical data analysis tasks, e.g., parameter prediction, model building, and theoretical verification. It involves multiple data modalities, including tabular data, images, and time series, and spans a broad range of sub-fields, including radio astronomy, time-domain astronomy, spectroscopy, stellar physics, astrophysics, galactic astronomy, cosmology, and astrometry. It also encompasses a diverse set of task types common in astronomical data analysis, such as classification, regression, clustering, model fitting, astrometric analysis, and theorem verification. The process of constructing NL2Astro is elaborated in Section 4.1. Especially, detailed descriptions of all tasks in NL2Astro are presented in Table 4.

Evaluation Metrics: To assess the effectiveness of AutoAstro, we adopt the following evaluation metrics: (i) *Accuracy*, which is the ratio of the number of correctly classified instances to the total number of test instances, to evaluate the performance of a classification model built by AutoAstro; (ii) *Root Mean Square Error (RMSE)*, which measures the errors between predicted and ground-truth values to evaluate the fitting quality of time-series and regression models built by AutoAstro, and *Normalized RMSE (NRMSE)*, which normalizes RMSE across different scales for fair comparison; (iii) *Aggregate Score (AS)*, which integrates Accuracy and NRMSE scores into a unified range of $[0, 10]$ to evaluate the overall performance of AutoAstro across all tasks; (iv) *Field Coverage Rate (FCR)* and *Redundant Field Rate (RFR)*, which evaluate the ratios of necessary and redundant fields used by a built model in the dataset to evaluate AutoAstro’s ability to understand the relationship between data and tasks; (v) *Average Time Cost (TiC)* and *Average Token Cost (ToC)*, which measure the efficiency of AutoAstro in completing tasks in terms of running time and token consumption; and (vi) *Pass Rate (PR)*, which evaluates the ratio of the number of successfully executed tasks to the number of tasks in NL2Astro. Detailed definitions and computations of these metrics are provided in Section 4.1.

Baselines: To the best of our knowledge, our work is the first LLM-based automated data analysis framework for astronomical tasks. We selected the following two

LLM-based frameworks for (general) data analysis as baselines: (i) *Agent Poirot* [29] and (ii) *DS-Agent* [30]. However, we note that Agent Poirot is designed for machine learning tasks and lacks the capacity for deep learning or temporal modeling. Consequently, our analysis focuses exclusively on its machine learning components. DS-Agent supports only modeling-related tasks and is therefore compared only for these tasks. In addition, DS-Agent imposes strict input requirements: users must manually define the dataset split and the evaluation code. To ensure compatibility, we adapted the data format according to its requirements. We then recruited 30 (iii) *human researchers* to participate in the evaluation. Each researcher was randomly assigned a specific task from NL2Astro. During task execution, participants were permitted to use an LLM for assistance. They were instructed to record the LLM’s token consumption and to document all interactions with the model — both inputs and outputs — in Markdown files. Researchers were required to complete the assigned tasks within a continuous, compact time frame and to provide a self-assessment of their task performance. The evaluation assessed whether the results included the necessary visualizations, explanatory information, and indicators. Self-assessments were reported on a 0–10 scale, where 0–3 indicated dissatisfaction, 4–7 indicated acceptable performance, and 8–10 indicated high satisfaction. The detailed experimental guidelines for researchers were presented in Fig. 3.

Environment: All experiments were conducted on a server equipped with an Intel(R) Core(TM) i5-12500 CPU, an NVIDIA(R) L20 GPU, and 32GB of memory, running Windows(R) 11. Meanwhile, Gemini-2.0-Flash¹, DeepSeek V3², Qwen-Max³, and GPT-4o⁴ were tested as base LLMs and accessed via their official APIs. By default, Gemini-2.0-Flash and DeepSeek V3 were used as base LLMs for image and non-image tasks, respectively.

2.2.2 Performance of AutoAstro with Different Base LLMs

To investigate the performance of AutoAstro with different base models, we selected three LLMs, namely DeepSeek, Qwen-Max, and GPT-4o, for evaluation. As shown in Fig. 4, we presented the performance of AutoAstro with these models on the NL2Astro dataset. In terms of Pass Rate, AutoAstro with DeepSeek V3 achieved the highest performance, reaching an overall rate of 86% across five repeated runs. We observed fluctuations in AutoAstro’s performance on time-series tasks when using Qwen-Max. Further analysis indicates that the instability stems from Qwen-Max’s tendency to construct complex neural network architectures or training schemes when dealing with time series with multiple input fields. This behavior suggests that Qwen-Max may exhibit limited flexibility in specific scenarios. In terms of Aggregate Score, AutoAstro with DeepSeek V3 again demonstrated the best overall performance. Tasks involving image analysis showed high scores across three task types because they are better supported by existing LLMs during pre-training. In terms of two efficiency indicators, Time Cost (in minutes) and Token Cost, AutoAstro with DeepSeek V3 also demonstrated clear advantages. It completed tasks in much less time, typically

¹<https://ai.google.dev/gemini-api>

²<https://api-docs.deepseek.com/>

³<https://qwen.ai/apiplatform>

⁴<https://openai.com/index/gpt-4o-system-card/>

Experimental Guidelines

Objective: Based on the randomly assigned task, you may use **code implementation**, **existing frameworks**, or other methods to achieve the **task goal**. During the process, you are encouraged to use **large language models (LLMs)** for assistance.

Metrics to Record (fill in the experiment log):

1. **Task Duration:** Record the time taken from starting the task to achieving the goal. Ensure the task is completed within a continuous and focused time period.
2. **LLM Usage:** You may use an LLM (*DeepSeek-V3*) for assistance. Record token consumption and save all interaction inputs and outputs in a Markdown file for later token calculation.
3. **Performance Metrics:** Record key performance indicators such as *Accuracy*, *RMSE*, etc.
4. **Self-Evaluation:** Assess your work based on:
 - Inclusion of necessary visualizations;
 - Quality of result interpretation;
 - Reporting of key metrics.
5. **Fields Used:** List all field names from the dataset that were used to solve the task.
6. **Final Rating (0–10):**
 - 0–3: Unsatisfactory;
 - 4–7: Acceptable;
 - 8–10: Excellent.
7. **Result Files:** Include the original code, intermediate results, and visualizations. Compress them into a ZIP file named `Dataset-TaskID-ResponsiblePerson`.

Notes:

- Ensure the task process (from start to completion) occurs within a tight and continuous timeframe.
- Tasks may involve *image classification (deep learning)*, *time series analysis (deep learning)*, *statistical analysis*, or *machine learning*.
- For cross-identification tasks, use the `Xmatch` Python package for data merging.
- For image classification, time series forecasting, or classification tasks, ensure training is done using only one A6000 or L20 GPU.

Fig. 3 Experimental guidelines for evaluation. For a fair comparison with automated LLM-based pipelines, we encourage human researchers to use LLMs in their tasks.

236 within 10 minutes. Furthermore, it achieved comparable or better performance while
237 incurring the lowest token costs. Finally, we presented two additional indicators that
238 reflect a model’s ability to align with task requirements: Average Field Coverage Rate
239 (Avg. FCR) and Average Redundant Field Rate (Avg. RFR). When DeepSeek V3 and
240 Qwen-Max were used, AutoAstro achieved high field coverage, effectively capturing
241 the fields needed for each task. In contrast, GPT-4o tended to “overthink,” introduc-
242 ing extraneous reasoning steps that sometimes diverted the task from its intended
243 direction, thereby reducing field utilization. While DeepSeek V3 achieved the highest
244 RFR, its Avg. RFR remained below 0.15, indicating negligible redundancy. In sum-
245 mary, across all evaluation dimensions, DeepSeek V3-based AutoAstro emerged as

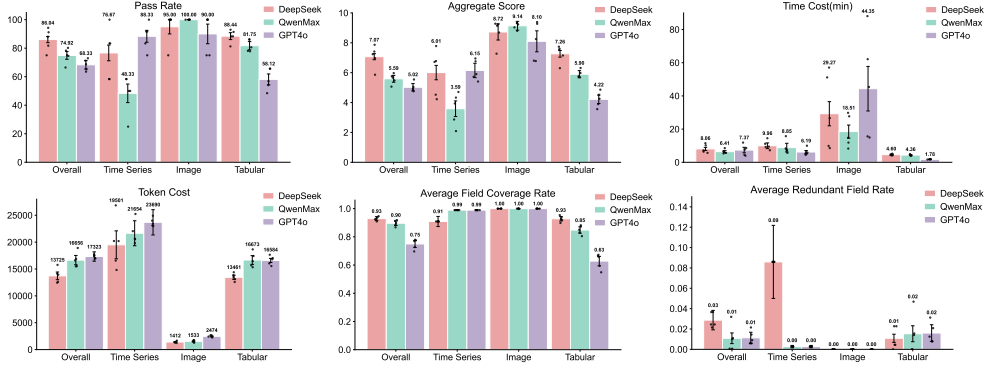


Fig. 4 Performance of AutoAstro with different base LLMs. We present the Pass Rate, Aggregate Score, Time Cost, Token Cost, Average Field Coverage Rate, and Average Redundant Field Rate of AutoAstro when using DeepSeek, Qwen-Max, and GPT-4o as base LLMs. We report the mean values from five repeated runs, with error bars indicating the standard deviation.

Table 1 Comparison of each LLM framework with human experts. ↑ and ↓ indicate higher and lower values denote better performance, respectively. The values in parentheses indicate the performance improvement or decline relative to human experts, with red indicating a decline and green indicating an improvement.

Method	AS↑	FCR↑	RFR↓	TiC↓	ToC↓	PR↑
Human Expert	7.62	0.97	0.07	82.67	11,509	100%
Agent Poirot	0.57 (↓92.52%)	0.26 (↓73.2%)	0.41 (↑485.71%)	1.88 (↓97.73%)	27,522 (↑139.13%)	88%
DS-Agent	5.23 (↓31.36%)	0.94 (↓3.09%)	0.61 (↑771.43%)	1.66 (↓97.99%)	2,210 (↓80.8%)	20%
AutoAstro	8.13 (↑6.69%)	0.92 (↓1.51%)	0.02 (↓71.43%)	8.32 (↓89.93%)	11,788 (↑2.42%)	91%

the most efficient, cost-effective, and high-performing model; thus, it was used by default in subsequent experiments.

2.2.3 Performance Comparison of AutoAstro vs. Baselines

We provide a comprehensive overview of each method’s performance. Fig. 5 presents the execution statuses of NL2Astro tasks by AutoAstro, Agent Poirot, and DS-Agent. Note that human experts successfully completed all randomly assigned tasks, thereby confirming the feasibility of the task design. Over the five runs, Agent Poirot failed to execute any image-related tasks. DS-Agent failed to perform any tasks other than modeling tasks and exhibited subpar performance even on modeling-related tasks. Compared with them, AutoAstro demonstrated broader task generalization across diverse types, exhibiting significantly wider applicability than Agent Poirot and DS-Agent.

Then, the overall performance of each method on each measure is shown in Table 1. In terms of AS, human experts achieved a high average score of 7.62, indicating strong confidence in completing each task. AutoAstro outperformed human experts, improving the AS score by 6.69%. DS-Agent exhibits a 31.36% performance reduction compared to human experts. Agent Poirot exhibited significantly poorer performance,

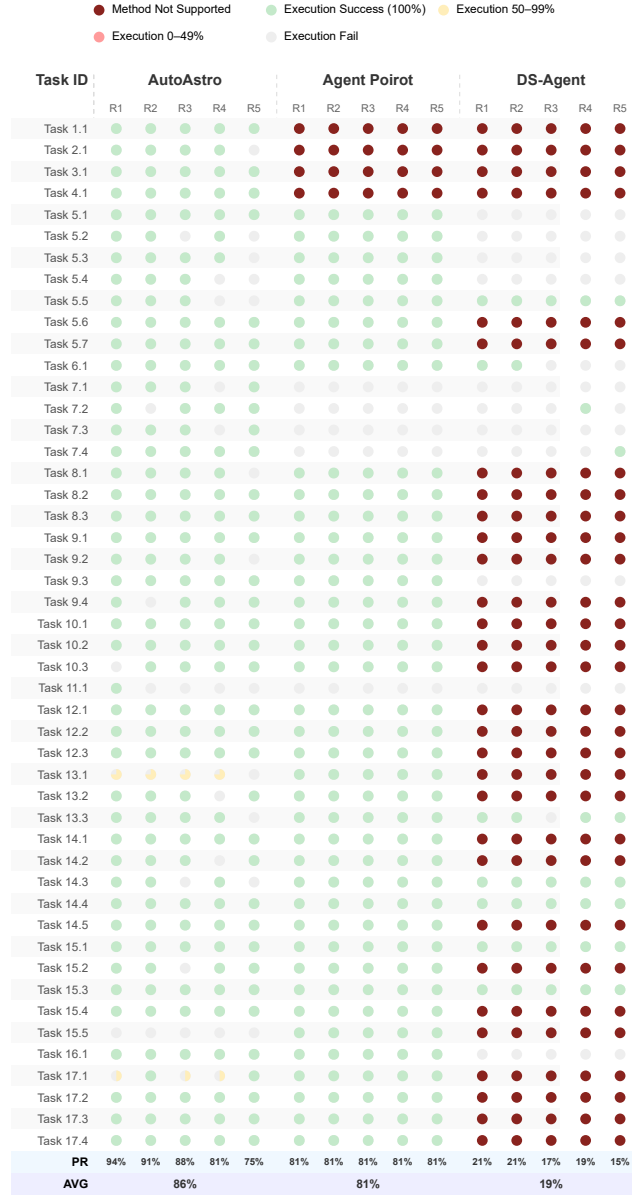


Fig. 5 Overall execution statuses of NL2Astro tasks. The figure shows the progress of different LLM frameworks in executing each task over five runs (R1–R5).

263 with a 92.52% degradation relative to human experts. In terms of FCR, human experts
 264 demonstrated superior performance, with lower field omission rates. Compared with
 265 human experts, AutoAstro showed only a 5.15% decrease in FCR. While DS-Agent

Table 2 Comparison with the performance of AutoAstro with that of existing methods.

Note that we report only the best performance for each task across all models with publicly available implementations in the literature.

Task ID	Method	Result	Metric
1.1	CGRU [38]	(burst) 96.8 & 0 (non burst) 99.5 & 1.5 (calibration) 99.9 & 0.3	TPR & FPR
	AutoAstro (VGG16)	(burst) 97.4 & 0.3 (non burst) 100 & 0 (calibration) 99.7 & 1.4	
2.1	CNN [39]	79%	Accuracy
	AutoAstro (Swinv2-small)	78%	
3.1	PICS-RESNET [40]	98.9%	Accuracy
	AutoAstro (Swin Transformer-Base)	99.1%	
4.1	H-CCNN [41]	94.8%	Accuracy
	AutoAstro (ResNet50)	97.6%	
6.1	LightGBM [35]	87%	Accuracy
	MLLM [35]	95%	
	AutoAstro (LSTM)	93%	
7.3	LightGBM [42]	(Teff)250.51, (logg)0.30, ([Fe/H])0.21	RMSE
	ResNet50 [42]	(Teff)452.8, (logg)0.25, ([Fe/H])0.17	
	GRU [42]	(Teff)440.31, (logg)0.34, ([Fe/H])0.31	
	AutoAstro (RNN)	(Teff) 219.78, (logg) 0.20, ([Fe/H]) 0.14	

achieves an FCR that is only 3.09% lower than that of human experts, it incurs an RFR that is seven times higher, resulting in significant computational inefficiency. Agent Poirot performed worse on both FCR and RFR compared to AutoAstro. Moreover, LLM-based automated frameworks demonstrate significant advantages over human experts in execution efficiency. Specifically, while human experts require an average of more than 1 hour per task, Agent Poirot, DS-Agent, and AutoAstro complete each task within 10 minutes, achieving approximately a 90% improvement. Regarding resource consumption, human experts (with LLM assistance) use approximately 10,000 tokens per task. Agent Poirot doubles token consumption, whereas AutoAstro maintains token consumption comparable to that of human experts. Although human experts achieve 100% PR, AutoAstro occasionally fails on specific task types. Nevertheless, it still significantly outperforms DS-Agent, with a PR of only 20%.

Furthermore, Table 2 presents the performance of models built by AutoAstro compared with the existing literature on several tasks. We observe that AutoAstro achieves performance comparable to, or even surpassing, that of state-of-the-art methods on both astronomical classification and physical parameter prediction tasks.

2.2.4 Error and Correction Analysis

Fig. 6 illustrates the distribution of error types that occurred during the execution of AutoAstro, as well as the results of automatic correction. We find that approximately 43.52% of execution errors were successfully resolved through the automatic correction mechanism. About 44.62% of tasks that would have otherwise failed were executed successfully after correction. In terms of error types, value misuse (Value

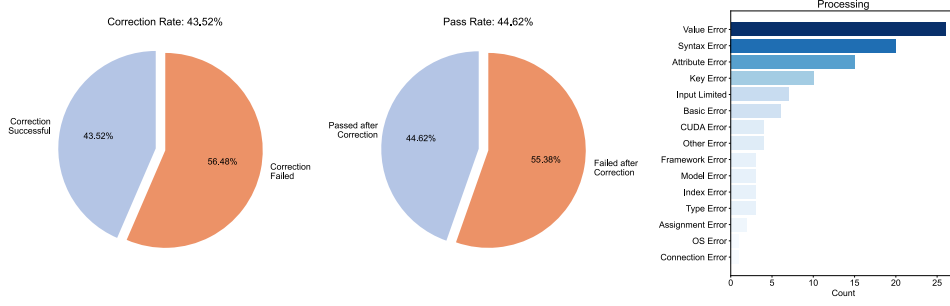


Fig. 6 Statistics of errors and corrections. The first two charts display the correction rate and the pass rate of tasks after error correction. The third chart presents the count of each error type during execution.

Error), improper syntax (Syntax Error), incorrect method usage (Attribute Error), and invalid field references (Key Error) were particularly prevalent.

2.3 Case Study

To further validate the applicability of AutoAstro, we present its performance on three real-world scientific tasks compared to that of human experts, Agent Poirot, and DS-Agent, including: (i) validating the inverse square law of brightness, (ii) constructing a spectral classification model, and (iii) predicting stellar parameters.

2.3.1 Validating the Inverse Square Law of Brightness

We illustrate Task 14.1, whose goal is to *investigate the relationship between absolute magnitude and luminosity to validate the inverse square law of brightness*, in Fig. 7.

The first part was completed by a human expert. It consists of four plots: they show the relationship between absolute magnitude (Mv) and luminosity, revealing a negative correlation. A linear regression was performed, yielding a slope of approximately -2.5948, which closely matches the theoretical value of -2.5. The final conclusion stated that “the logarithmic relationship between absolute magnitude and luminosity has been validated,” with an R^2 value of 0.9537, indicating a high degree of model fit. The second part was carried out by AutoAstro. Its plots also revealed the relationship between luminosity and absolute magnitude, initially suggesting an inverse relationship. During the data analysis phase, correlations were assessed using both Pearson and Spearman coefficients. AutoAstro derived a slope of -2.595, which is also nearly identical to the theoretical value. Ultimately, its data analysis and visualization supported the same conclusion as the manual approach: there is a negative correlation between absolute magnitude and luminosity. The third part was performed by Agent Poirot. However, its analysis deviated from the task’s original objective. Instead, it focused on answering the following question: “How does the luminosity of stars vary across different spectral classes, and are there any outliers or anomalies in this relationship?” As a result, it did not directly address the specific relationship between absolute magnitude and luminosity. These cases demonstrate that AutoAstro has strong data processing and pattern recognition capabilities. In tasks with well-defined objectives

and data, it can, to a considerable extent, serve as a substitute for manual analysis, particularly in large-scale data exploration and quantitative validation. However, we also note that AutoAstro’s effectiveness depends on clear task definitions. For complex analyses requiring domain-specific insight, human expertise remains indispensable. Finally, DS-Agent failed on this task due to its lack of tool-use capabilities.

2.3.2 Constructing a Spectral Classification Model

The goal of Task 14.3 is to predict stellar spectral classes. Specifically, it requires building a classification model using the features ‘Temperature (K)’, ‘Luminosity (L/L_o)’, ‘Radius (R/R_o)’, and ‘Absolute magnitude (M_v)’ to predict the ‘Spectral Class’ field, as shown in Fig. 8.

This is a representative real-world scenario involving heterogeneous stellar catalogs that include main-sequence stars, giants, and dwarfs, in which spectral classification provides valuable physical insights into these objects. As a baseline, a manually configured Random Forest (RF) model achieves an accuracy of 0.8542. However, it exhibits notable confusion between early-type stars (i.e., O, B, A classes). In contrast, AutoAstro not only achieves higher accuracy (0.875) but also provides interpretability via feature importance analysis and ROC curves, offering greater transparency for manual inspection. The feature importance analysis reveals that temperature is the most influential predictor for spectral classification—a result consistent with astrophysical expectations. Despite its improved performance, AutoAstro still exhibits some misclassification, particularly between early- and late-type stars, with increased confusion around K-type stars. For comparison, DS-Agent also employs a Random Forest model but provides no interpretability details and achieves lower accuracy than the manual approach. Finally, Agent Poirot failed on this task due to the absence of the required tools.

2.3.3 Stellar Parameters Prediction

We further consider Task 7.3, whose goal is to *build a time-series prediction model using all flux data to simultaneously forecast the effective temperature (‘Teff’), surface gravity (‘logg’), and metallicity (‘[Fe/H]’) fields*, in Fig. 9.

We evaluate a stellar parameter regression task using stellar spectra from LAMOST DR10. This scenario is common in astrophysics, where deep learning models are employed to infer stellar parameters. The manually constructed model was developed by a non-expert in astronomy, resulting in an architecture that fails to capture meaningful patterns in the data. Consequently, the model tends to regress all predictions toward the mean or median of the target distributions, thereby minimizing mean squared error (MSE) without learning genuine spectral–parameter relationships. This behavior is further supported by a non-decreasing training loss curve. Such issues are commonly observed when novice practitioners, who are unfamiliar with the domain-informed dataset, attempt to apply ML models without a sufficiently specific design. AutoAstro significantly outperforms manual models for this task. Although LSTM networks are typically more suited to time-series data (e.g., light curves), AutoAstro adapts them for spectral regression and demonstrates effective learning throughout

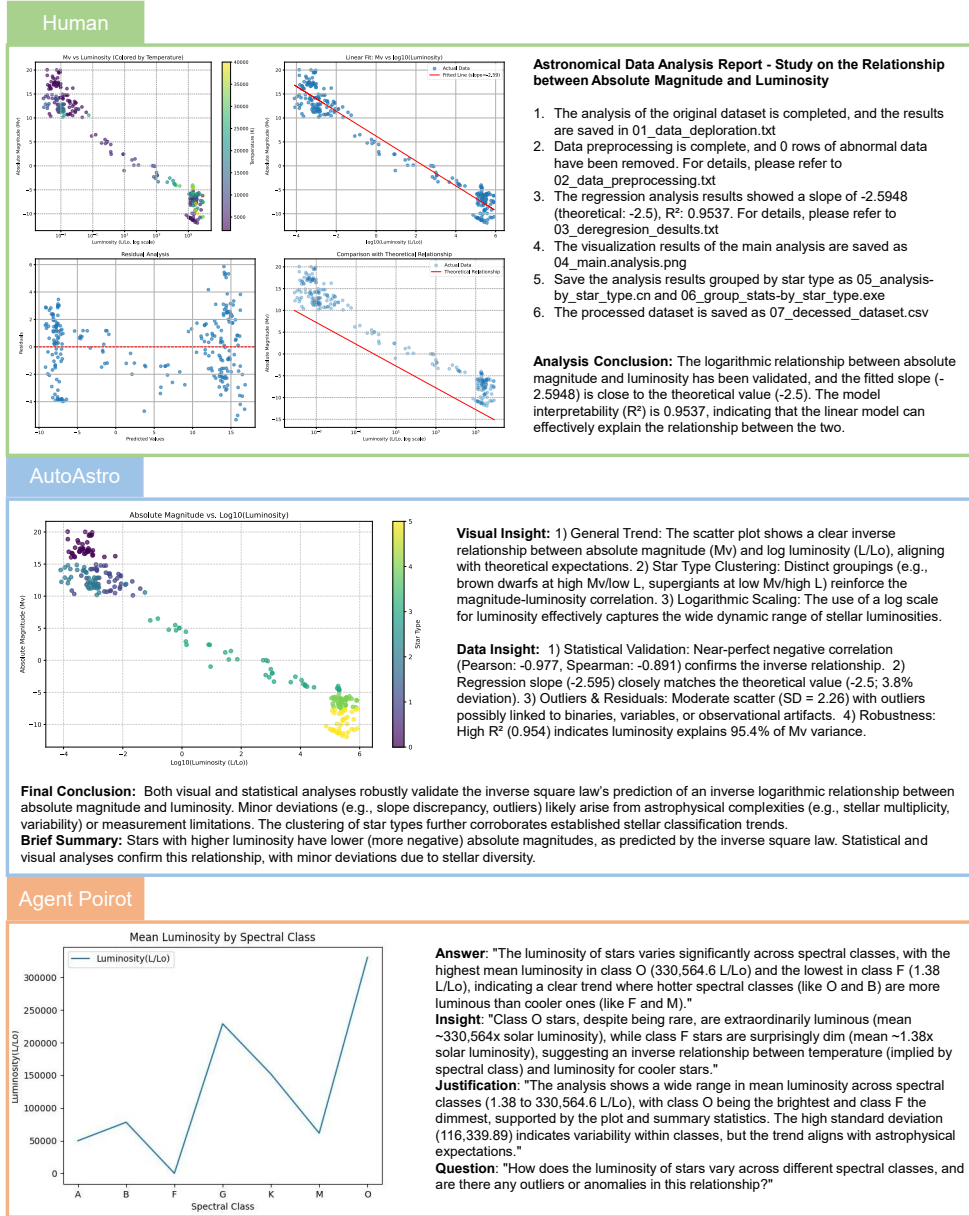


Fig. 7 Outputs for validating the inverse square law of brightness. We present the results by human researchers, AutoAstro, and Agent Poirot.

359 training. The loss continues to decrease over epochs, suggesting potential for further
 360 improvement with extended training or architectural refinement. The model achieves
 361 an RMSE of approximately 220K in Teff, 0.20 dex in logg, and 0.13 dex in [Fe/H],
 362 performance levels that are reasonable for a preliminary study and methodological

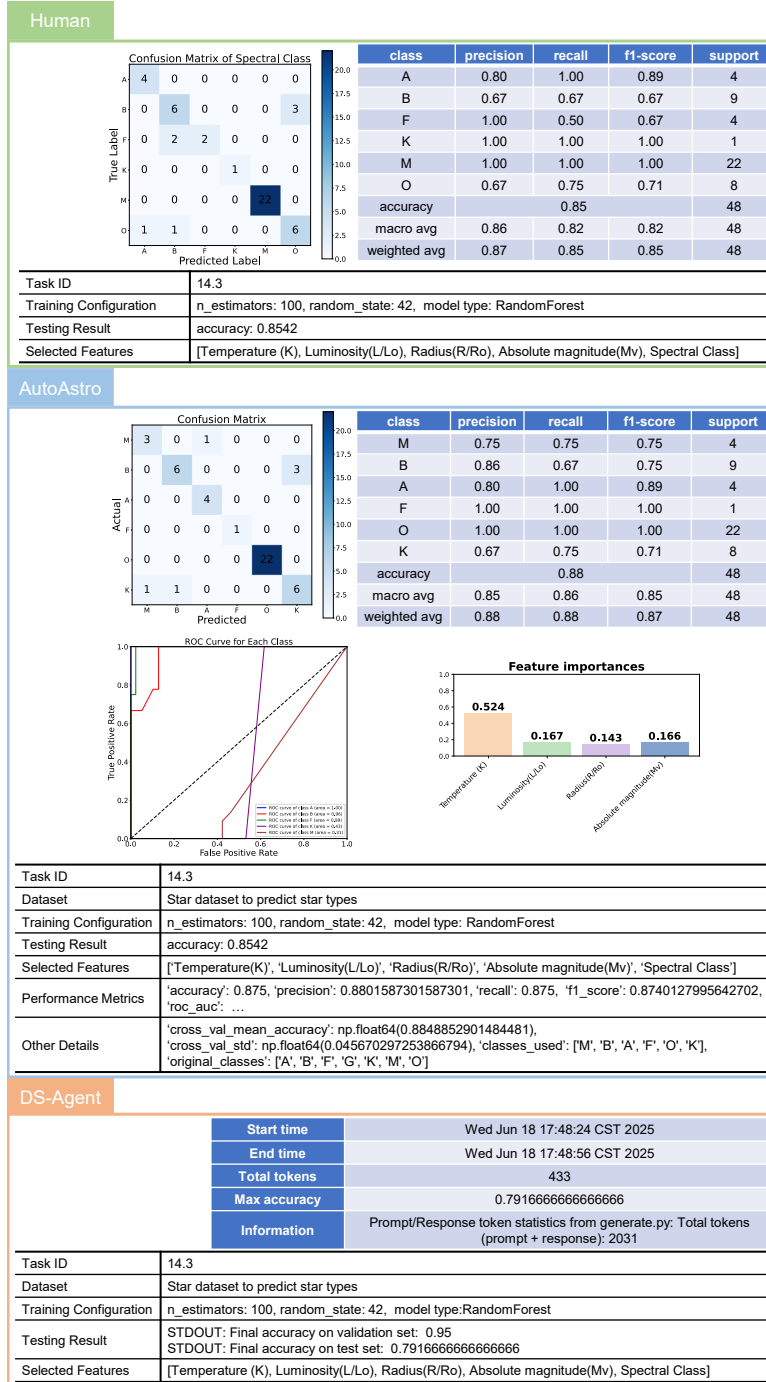


Fig. 8 Outputs for constructing a spectral classification model. We present the results by human researchers, AutoAstro, and DS-Agent.

363 exploration. Finally, Agent Poirot and DS-Agent both failed on this task because they
364 were unable to build regression models.

365 3 Discussion

366 We introduce NL2Astro, the first comprehensive benchmark dataset for astronomi-
367 cal data analysis tasks, and develop AutoAstro, an LLM-empowered framework for
368 automated data analysis workflow construction. NL2Astro covers a diverse set of
369 representative tasks, along with their corresponding data processing and model-
370 ing workflows. For each task, we define explicit and quantifiable evaluation metrics,
371 thereby providing a unified benchmark for studying and comparing automated
372 astronomical data analysis systems. Built upon this dataset, AutoAstro automati-
373 cally performs early-stage exploratory analysis procedures commonly encountered in
374 astronomical research, including data preprocessing, feature engineering, ML model
375 training, and visualization. Experimental results show that AutoAstro improves the
376 efficiency of astronomical data analysis, reducing preliminary exploration that can
377 take hours or more when performed manually to just a few minutes. This efficiency
378 gain allows researchers to spend less time on repetitive operations and devote more
379 effort to formulating scientific questions, interpreting physical mechanisms, and
380 conducting reasoning and experimental design.

381 Beyond improving efficiency, AutoAstro also lowers, to some extent, the technical
382 barrier to astronomical data analysis. Conventional astronomical research workflows
383 often require researchers to possess strong programming skills, data engineering
384 experience, and domain expertise simultaneously. By automating data processing and
385 analysis pipelines, AutoAstro enables researchers with only astronomy backgrounds
386 to participate in complex data-driven studies. This capability not only enhances the
387 accessibility of data-intensive astronomical research but may also reshape future
388 modes of scientific collaboration and knowledge production by facilitating method-
389 ological sharing, intellectual exchange, and interdisciplinary innovation among
390 researchers with diverse backgrounds.

391 From a system design perspective, AutoAstro is built on top of pretrained LLMs
392 (both open- and closed-source), a Python interpreter, and neural network frame-
393 works. It can therefore be viewed as a human-AI collaborative framework tailored
394 to scientific scenarios. By adapting the capabilities of underlying foundation models,
395 AutoAstro can not only perform astronomy-related data processing and analysis tasks
396 but also provide general-purpose functionalities such as question answering, code
397 generation, multi-step reasoning, and summarization. At the same time, AutoAstro
398 adopts an open and modular software architecture, in which functional components
399 are decoupled through unified interfaces. This design allows the system to flexibly
400 integrate new foundation models, scientific computing tools, and domain-specific
401 analysis modules. Such an architecture improves the system’s maintainability and
402 extensibility while enabling researchers to continually benefit from rapid advances
403 in the LLM ecosystem and open-source AI toolchains. As a result, AutoAstro has the
404 potential to adapt and evolve toward increasingly complex scientific tasks.

405 It is important to emphasize that AutoAstro is not intended to replace researchers’
406 scientific judgment, nor is it designed to independently generate publishable sci-
407 entific findings. Instead, it is positioned as an assistant for early-stage exploration,
408 model construction, and data analysis within the scientific workflow. Accordingly,
409 the case studies primarily focus on exploratory tasks that researchers routinely per-
410 form to reflect the processes of problem analysis and validation in the early stages of
411 scientific research. Although the tasks included in NL2Astro are not highly complex
412 scientific reasoning problems in themselves, these cases demonstrate AutoAstro’s
413 capabilities in task localization, goal analysis, self-guided exploratory analysis, and
414 rapid result validation. Under the traditional research paradigm, such work typically
415 requires extensive manual operations, including data retrieval, script writing, result
416 verification, and repeated tool switching. In contrast, through an interactive and iter-
417 ative human-AI collaboration mechanism, AutoAstro enables researchers to quickly
418 develop an initial understanding of a problem and efficiently complete preliminary
419 steps, such as candidate target selection, model construction, result validation, and
420 hypothesis testing. This significantly reduces both the cognitive and operational costs
421 of the exploratory stage.

422 At the same time, the multi-turn interactive research workflow exposes a central
423 challenge in current agentic systems for scientific problems: context management. The
424 system must not only maintain historical information relevant to the current research
425 objective but also dynamically adapt to changes in user intent, analysis trajectory, and
426 task goals throughout the interaction. In particular, for long-horizon and multi-stage
427 scientific workflows, selective context retention, state updating, and task memory
428 directly affect the reasoning stability and interaction efficiency of agentic systems. We
429 argue that context management remains an important open problem for LLM-based
430 agents and constitutes a key direction for the future evolution of AutoAstro.

431 Although we focus primarily on astronomy, the framework and methodology
432 embodied by AutoAstro have broader potential applicability. Core procedures such
433 as data integration, model invocation, automated analysis, and result interpretation
434 are not unique to astronomy; rather, they are widely present in life sciences, medicine,
435 and other data-intensive domains. By adapting to the data structures, task paradigms,
436 and specialized toolchains of different disciplines, agentic systems like AutoAstro can
437 be extended to a broader range of scientific scenarios and serve as early prototypes
438 for general-purpose AI research assistant frameworks.

439 The long-term effectiveness of such a system depends heavily on the breadth,
440 quality, and timeliness of the data sources and task types it can access. As new obser-
441 vational data, analysis methods, and scientific tasks continue to emerge, an important
442 future work is to continuously incorporate these changes into the system’s capability
443 boundary. For its open architecture, the research community can contribute new data
444 interfaces, analysis modules, and task templates, thereby driving the continuous evo-
445 lution of the system. Through community-driven iterative updates, AutoAstro may
446 gradually develop into an open, dynamic, and extensible intelligent infrastructure
447 for scientific research.

448 For complex AI scientific research systems such as AutoAstro, output stability
449 is also a critical issue. Current LLMs cannot guarantee deterministic outputs under

identical inputs, even when the temperature parameter is set to 0. AutoAstro naturally inherits this uncertainty from the underlying models. In addition, external analysis tools, data processing procedures, and visualization algorithms may further introduce run-to-run variations. It is therefore expected that multi-step scientific workflows may exhibit some variability across runs. As such, we report average results from five independent runs to assess the system’s consistency and to further investigate how prompt specificity affects workflow stability. The experimental results indicate that clearer task descriptions, more explicit methodological constraints, and more standardized workflow designs can effectively reduce output variability and improve system stability.

Notably, such run-to-run variability does not necessarily indicate a system flaw. To some extent, the distinct analytical trajectories produced across multiple runs resemble the methodological differences that may arise when researchers analyze the same data and task. This diversity can sometimes reveal alternative analysis strategies and expose implicit degrees of freedom in the research workflow. Therefore, during the early exploratory stage of scientific inquiry, moderate output diversity may not be a purely negative property, but rather a capability that can be strategically leveraged. Future systems may further develop toward an “explore-then-converge” research paradigm, in which candidate solutions are first generated over a broader solution space and then gradually refined toward more optimal, robust, or interpretable research pathways.

As AI plays an increasingly prominent role in scientific research, its potential ethical and societal implications must also be carefully examined. The automation of scientific tasks is gradually blurring the boundary between human contributions and machine-generated content and may pose new challenges to existing research norms concerning authorship, intellectual property, and scientific accountability. Moreover, the deployment of systems such as AutoAstro may have far-reaching consequences for the training of future researchers. On the one hand, such systems can lower the barrier to entry for scientific research and help new researchers engage with research workflows more quickly. On the other hand, excessive reliance on automated systems may weaken researchers’ fundamental analytical skills, problem modeling abilities, and critical thinking. In the long run, this trend may leave some researchers insufficiently equipped to verify, challenge, or improve AI-generated results, thereby affecting research quality and scientific innovation.

Furthermore, substantial disparities may exist across research communities and disciplines in access to AI tools, computational resources, and capacity for technical adoption. Such technological inequality may further widen gaps in research opportunities and scientific output. Meanwhile, as AI systems become deeply involved in the scientific process, they may also indirectly shape research directions. For example, researchers may become more inclined to prioritize problems that are easier for systems such as AutoAstro to process while overlooking research questions that are not yet amenable to automated analysis but nevertheless possess significant scientific value. In extreme cases, this tendency may lead to the convergence of research topics, reduced methodological diversity, and a potential imbalance in the scientific ecosystem.

495 In light of these considerations, the design and deployment of AI systems for
496 science, such as AutoAstro, should proceed in a cautious and responsible man-
497 ner. Although improving research efficiency is important, the central role of human
498 researchers in scientific discovery remains irreplaceable. The human-AI collaborative
499 paradigm emphasizes that AI should be regarded as an assistive tool in the research
500 process, rather than an independent scientific agent. The formulation of key scientific
501 questions, the judgment of research directions, the interpretation of results, and the
502 assumption of final responsibility should remain with human researchers. As these
503 technologies continue to advance, the research community should gradually estab-
504 lish guidelines and best practices for AI-assisted scientific research, with the goal of
505 clarifying boundaries of responsibility, preserving research integrity, and promoting
506 transparent, interpretable, and auditable mechanisms for human-AI collaboration.
507 Only by striking an appropriate balance between efficiency gains and scientific norms
508 can the potential of AI-assisted scientific discovery be realized in a sustainable and
509 responsible manner.

510 4 Method

511 4.1 NL2Astro Benchmark Construction

512 To evaluate the data analysis capabilities of AutoAstro, we constructed a natural
513 language instruction dataset named NL2Astro. Next, we outline the dataset collection
514 process and the evaluation protocol.

515 4.1.1 Data Collection

516 As shown in Table 3, the datasets in NL2Astro are collected from the following three
517 sources: (i) public datasets, (ii) Kaggle datasets, and (iii) observation databases. Next,
518 we will describe in more detail how these datasets are obtained and processed.

519 **Public Datasets:** Several methodological studies curate and publish domain-
520 specific datasets to validate their methods. In this study, we collected publicly
521 available data reported in [39–41, 43]. These data sources provide ground-truth labels,
522 detailed contextual metadata, and scientifically relevant features, which are essential
523 for training and evaluating advanced models for astrophysical analysis.

524 **Kaggle Datasets:** Kaggle⁵ is one of the main platforms for data science competi-
525 tions, hosting an extensive repository of technical reports and code contributed by
526 a global community of experienced data scientists. We collected astronomical data
527 available on Kaggle and performed preprocessing to address potential issues with
528 noisy or inconsistent data.

529 **Observation Databases:** The existing observation databases contain publicly
530 accessible time-domain data from major astronomical surveys and mission archives.
531 In this study, we utilize datasets from two surveys: LAMOST [2] and Kepler [44].

532 We confirm that all datasets used do not involve any copyright infringement and
533 are free to use and distribute under the CC-BY-4.0 license.

⁵<https://www.kaggle.com/>

Table 3 List of datasets used to construct the NL2Astro benchmark.

Dataset	Source	Modality	Size
SpectrumCls	Public	Image	8,815
RadioGalaxyDataset	Public	Image	7,158
HTRU_Medlat	Public	Image	11,195
FAST_2018	Public	Image	15,482
PLAsTiCC	Kaggle	Time Series	7,848
Kepler_Class	Observation Database	Time Series	447,404
LAMOST DR10	Observation Database	Spectrum	99,597
Bright Star	Public	Tabular	516
celestial_object_classify	Kaggle	Tabular	100,000
Gaia Astronomical Data	Kaggle	Tabular	2,557
Kepler labelled time series	Kaggle	Time Series	5,657
simulated Hubble’s Law Data	Kaggle	Tabular	400
Solar Flares from RHESSI Mission	Kaggle	Tabular	116,143
Star types	Kaggle	Tabular	240
SDSS17	Kaggle	Tabular	100,000
Light curve of a variable star	Observation Database	Time Series	22,372
Tycho Astrometry Catalog	Kaggle	Tabular	2539913

4.1.2 Question Type

The main goal of NL2Astro is to assess the performance of LLM-based agentic frameworks in astronomy. To achieve this goal, we convert standard datasets into natural language instructions. Therefore, as shown in Tables 4 and 5, we manually constructed 48 natural language instructions tailored to the characteristics of each dataset. The design of these tasks aims to comprehensively evaluate AutoAstro’s capabilities in handling astronomical data across various domains and data types.

4.1.3 Evaluation Protocol

We introduce all metrics used in NL2Astro to quantitatively assess task completion status. Specifically, *accuracy* is employed to measure classification performance, and *root mean square error* (RMSE) is used to evaluate the performance of parameter estimation; For human judgment, we introduce scoring functions that assign scores from 1 to 10, with higher values indicating better performance. Time cost (TiC) and token cost (ToC) represent the execution time of each task and the token consumption incurred when leveraging LLMs, respectively. Then, $\overline{TiC}$ and $\overline{ToC}$ represent the average values of TiC and ToC across all tasks.

Then, we introduce the compound metrics used to evaluate the overall performance of each method. The normalized RMSE (NRMSE) extends RMSE to address its sensitivity to variations in the scale of the estimated values, which is defined in Eq. (2), with Ω denoting the set of all tested RMSE values:

$$\text{NRMSE} = \frac{\text{RMSE}}{\max(\Omega) - \min(\Omega)}. \quad (2)$$

The aggregate score (AS) harmonizes results across diverse evaluation metrics and assesses each method’s overall performance. As shown in Eq. (3), AS integrates

Table 4 Descriptions of tasks in NL2Astro.

ID	Task Description
1.1	Build a solar eruptions classification model
2.1	Build a radio galaxy morphology classification model
3.1 & 4.1	Build a pulsar candidate classification model
5.1	Use flux data to build a time-series classification model with the target field
5.2	Use flux data to build a time-series prediction model for forecasting the 'mwebv' field
5.3	Use flux data to build a time-series classification model with the 'ddf' field
5.4	Classify astronomical sources using flux measurements and redshift data
5.5	Predict the spectroscopic redshift using photometric redshift and flux measurements
5.6	Cluster astronomical sources based on infrared magnitudes to identify distinct groups of objects
5.7	Explore the relationship between host galaxy redshift and infrared magnitudes
6.1	Use flux data to build a time-series classification model for the 'Label' field
7.1	Use flux data to build a time-series classification model for the 'Type' field
7.2	Use flux data to build a time-series forecasting model for the 'Teff' field.
7.3	Build a time-series forecast model using flux data for the 'Teff', 'logg', and '[Fe/H]' fields.
7.4	Build a time-series classification model using 'Teff', 'logg', and '[Fe/H]' to predict the 'Type' field.
8.1	Use right ascension and declination to draw a mollwide diagram
8.2	Create a 2D heatmap of RA/Dec binned by median vmag and identify outliers in vmag for potential anomalies or errors
8.3	Investigate the relationship between vmag and infrared magnitudes
9.1	Examine spatial distribution of objects using right ascension and declination on a mollwide diagram, and be careful not to perform any sampling during the analysis process.
9.2	Compare Petrosian radii and fluxes across classes to study object morphologies
9.3	Train a classification model using photometric features and evaluate its performance
9.4	Investigate correlations between Petrosian fluxes and radii within each class
10.1	Visualize the spatial distribution of object types in the galactic plane using longitude and latitude
10.2	Investigate correlations between vertical and radial distances to test for systematic trends
10.3	Analyze the vertical distribution of objects to identify trends or clustering relative to the galactic plane
11.1	Use all flux data to build a time-series classification model for the 'LABEL' field
12.1	Investigate the relationship between apparent magnitude and redshift to understand how galaxy distance correlates with velocity
12.2	Analyze the distribution of redshift values to identify any anomalies or clustering that might indicate peculiar velocities or measurement biases
12.3	Convert apparent magnitude to luminosity distance and plot against redshift to visualize the Hubble relationship
13.1	Classify flares based on their duration and peak count rates to identify distinct temporal patterns
13.2	Predict flare duration based on peak count rates and energy range
13.3	Train and evaluate a logistic regression model predicting peak.c/s (binned into categories) using duration.s and temporal features
14.1	Investigate the relationship between absolute magnitude and luminosity to validate the inverse square law of brightness
14.2	Plot stars on the HR diagram (Temperature vs. Luminosity) to visually identify outliers that do not conform to expected patterns
14.3	Build a classification model using 'Temperature (K)', 'Luminosity(L/Lo)', 'Radius(R/Ro)', and 'Absolute magnitude(Mv)' to predict the 'Spectral Class' field
14.4	Build a classification model using 'Temperature (K)', 'Luminosity(L/Lo)', 'Radius(R/Ro)', and 'Absolute magnitude(Mv)' to predict the 'Star Type' field
14.5	Use clustering to group stars based on Temperature, Luminosity, and Radius, and identify points far from any cluster
15.1	Train a logistic regression model to predict 'class' using photometric features and redshift
15.2	Investigate how redshift affects the correlations between photometric bands (u-z) in quasars, to understand their evolution with distance and improve redshift estimation methods
15.3	Build a regression model to predict redshift using photometric bands (u-z) for quasars and evaluate its performance
15.4	Investigate positional error characteristics across different object types to assess measurement uncertainties
15.5	Analyze the distribution of metallicity ([Fe/H]) among stars to identify distinct populations (e.g., metal-poor vs. metal-rich stars)
16.1	Use flux data to build a time-series classification model for the 'Label' field
17.1	Analyze the distribution of proper motions to identify high-velocity stars or outliers
17.2	Examine the spatial distribution of stars using RAdeg and DEdeg to identify overdensities or voids
17.3	Investigate the relationship between BTmag and VTmag to classify stars by color and identify potential variable stars
17.4	Explore correlations between positional errors and epoch differences to assess temporal effects

Table 5 Details of tasks in NL2Astro. We use “Clean?” to indicate whether the dataset is curated, i.e., has undergone cleaning and preprocessing procedures, “Cross?” to indicate whether the dataset supports cross-matching with other datasets, for example, through the availability of positional information, “Subject” to indicate the subfield of astronomy to which the dataset belongs.

Task ID	Dataset	Clean?	Cross?	Subject	Task Type
1.1	SpectrumCls	✓	✗	Solar Radio Astronomy	Classification
2.1	RadioGalaxyDataset	✓	✗	Galaxy Morphology	Classification
3.1	HTRU_Medlat	✓	✗	Radio Astronomy	Classification
4.1	FAST_2018	✓	✗	Radio Astronomy	Classification
5.1	PLAsTiCC	✓	✓	Time-Domain Astronomy	Classification
5.2	PLAsTiCC	✓	✓	Time-Domain Astronomy	Regression
5.3	PLAsTiCC	✓	✓	Time-Domain Astronomy	Classification
5.4	PLAsTiCC	✓	✓	Time-Domain Astronomy	Classification
5.5	PLAsTiCC	✓	✓	Time-Domain Astronomy	Regression
5.6	PLAsTiCC	✓	✓	Time-Domain Astronomy	Classification
5.7	PLAsTiCC	✓	✓	Time-Domain Astronomy	Model Fitting
6.1	Kepler_Class	✓	✗	Time-Domain Astronomy	Classification
7.1	LAMOST DR10	✓	✗	Spectroscopy	Classification
7.2	LAMOST DR10	✓	✗	Spectroscopy	Regression
7.3	LAMOST DR10	✓	✗	Spectroscopy	Regression
7.4	LAMOST DR10	✓	✗	Spectroscopy	Classification
8.1	Bright Star	✓	✓	Stellar Physics	Astrometry
8.2	Bright Star	✓	✓	Stellar Physics	Astrometry
8.3	Bright Star	✓	✓	Stellar Physics	Model Fitting
9.1	celestial_object_classify	✓	✓	Astrophysics	Astrometry
9.2	celestial_object_classify	✓	✓	Astrophysics	Model Fitting
9.3	celestial_object_classify	✓	✓	Astrophysics	Classification
9.4	celestial_object_classify	✓	✓	Astrophysics	Model Fitting
10.1	Gaia Astronomical Data	✗	✗	Galactic Astronomy	Astrometry
10.2	Gaia Astronomical Data	✗	✗	Galactic Astronomy	Astrometry
10.3	Gaia Astronomical Data	✗	✗	Galactic Astronomy	Astrometry
11.1	Kepler labelled time series	✓	✗	Exoplanetology	Classification
12.1	simulated Hubble’s Law Data	✓	✗	Cosmology	Regression
12.2	simulated Hubble’s Law Data	✓	✗	Cosmology	Model Fitting
12.3	simulated Hubble’s Law Data	✓	✗	Cosmology	Theorem Verification
13.1	Solar Flares from RHESSI Mission	✗	✗	Solar Physics	Classification
13.2	Solar Flares from RHESSI Mission	✗	✗	Solar Physics	Regression
13.3	Solar Flares from RHESSI Mission	✗	✗	Solar Physics	Regression
14.1	Star types	✓	✗	Stellar Physics	Theorem Verification
14.2	Star types	✓	✗	Stellar Physics	Theorem Verification
14.3	Star types	✓	✗	Stellar Physics	Classification
14.4	Star types	✓	✗	Stellar Physics	Classification
14.5	Star types	✓	✗	Stellar Physics	Clustering
15.1	SDSS17	✓	✓	Stellar Physics	Model Fitting
15.2	SDSS17	✓	✓	Stellar Physics	Model Fitting
15.3	SDSS17	✓	✓	Stellar Physics	Regression
15.4	SDSS17	✓	✓	Stellar Physics	Astrometry
15.5	SDSS17	✓	✓	Stellar Physics	Clustering
16.1	Light curve of a variable star	✓	✗	Time-Domain Astronomy	Classification
17.1	Tycho Astrometry Catalog	✗	✓	Astrometry	Astrometry
17.2	Tycho Astrometry Catalog	✗	✓	Astrometry	Astrometry
17.3	Tycho Astrometry Catalog	✗	✓	Astrometry	Classification
17.4	Tycho Astrometry Catalog	✗	✓	Astrometry	Astrometry

different scores and normalizes them to a unified score in the range [1, 10]:

$$AS = \frac{1}{n} \sum_{i=1}^n \begin{cases} \frac{\text{score}_i + M_{\text{score}_i} \times 10}{2}, & \text{if } M_{\text{score}_i} \neq 0; \\ \text{score}_i, & \text{if } M_{\text{score}_i} = 0. \end{cases} \quad (3)$$

Here, M_{score_i} is either one minus NRMSE or accuracy, both in the range [0, 1].

From a task-oriented perspective, we define a set of standard fields for each task. Similarly, for each evaluation task, the fields used by a method are recorded to compute both the field coverage rate (FCR) and the redundant field rate (RFR). All n tasks have their standard field sets $S_1, S_2, \dots, S_n$, and T is the set of fields used by a method. Given the union of all standard fields $S_{all} = \bigcup_{i=1}^n S_i$, we define the FCR and RFR of a method in Eq. (4):

$$FCR = \frac{|T \cap S_{all}|}{|S_{all}|} \quad \text{and} \quad RFR = \frac{|T \setminus S_{all}|}{|T|}, \quad (4)$$

Here, $|\cdot|$ denotes the set cardinality operator.

4.2 AutoAstro Implementation

In this section, we present the detailed implementation of key modules in AutoAstro.

4.2.1 Data Cross-Matching Module

With the rapid growth of astronomical data from diverse sources, cross-matching datasets enables astronomers to build a comprehensive, multi-dimensional understanding of celestial objects. It allows for the identification of counterparts across different wavelengths, the verification of observations, the refinement of source properties, and the discovery of rare or transient phenomena. To facilitate the efficient integration of heterogeneous astronomical surveys, we developed an automatic cross-matching module based on the XMatch service.⁶ The module is designed to identify corresponding astronomical sources across large-scale catalogs while minimizing manual preprocessing and metadata curation. Specifically, given an input dataset description D_{desc} and a cross-match catalog directory $\text{Dir}_{\text{cross}}$, it first performs catalog recommendation by identifying candidate reference catalogs that are compatible with the uploaded dataset, i.e.,

$$\text{Dir}_{\text{suggested}} = \mathcal{R}(\text{Dir}_{\text{cross}}), \quad (5)$$

where $\mathcal{R}(\cdot)$ denotes the recommendation procedure. Users then select a target catalog D_{selected} from the suggested candidates. The module applies an LLM-based structural analysis to infer the fields suitable for positional cross-matching, i.e.,

$$F = \mathcal{L}(D_{\text{desc}}), \quad (6)$$

⁶<https://cds.unistra.fr/help/documentation/xmatch-more/api/>

583 where F represents the set of inferred matching-related fields, including positional
 584 coordinates and associated identifiers.

585 Cross-matching is then executed through the XMatch API using a positional
 586 matching radius of $\theta = 3''$, i.e.,

$$M = \mathcal{X}(D_{\text{desc}}, D_{\text{selected}}, F; \theta), \quad (7)$$

587 where M denotes the resulting matched dataset and $\mathcal{X}(\cdot)$ represents the cross-
 588 matching operation. Our implementation supports major astronomical catalogs (e.g.,
 589 II/246/out from the 2MASS Point Source Catalog, which contains approximately
 590 4.71×10^8 sources).

591 For each newly generated field, it automatically retrieves metadata descriptions
 592 and external references through a field-linking procedure, i.e.,

$$F_{\text{new}} = \mathcal{H}(D_{\text{desc}}, D_{\text{selected}}, F), \quad (8)$$

593 where $\mathcal{H}(\cdot)$ denotes automated metadata association. The final dataset is built as

$$D_{\text{new}} = \mathcal{U}(D_{\text{desc}}, M, F_{\text{new}}), \quad (9)$$

594 with $\mathcal{U}(\cdot)$ representing the dataset update operation.

595 Although the above workflow is fully automated in most cases, a small subset
 596 of catalog fields may lack sufficient metadata for automatic reference generation. In
 597 such cases, it provides a lightweight manual interface that allows users to supplement
 598 missing field descriptions.

599 4.2.2 Machine Learning Module

600 Machine learning (ML) is a principal paradigm for scientific data analysis and has
 601 been widely adopted in astronomical research for classification, prediction, anomaly
 602 detection, and knowledge extraction [15]. The proposed module is designed as an
 603 automated, multidimensional analytical framework that integrates visualization, sta-
 604 tistical inference, and ML-based modeling into a unified reasoning process. The
 605 framework emphasizes analytical consistency rather than isolated task execution,
 606 thereby enabling comprehensive interpretation of heterogeneous tabular datasets.
 607 Given a tabular dataset D_t and an analytical task T , the ML analysis process can be
 608 formulated as:

$$\mathcal{F}_{ml}(D_t, T) = \mathcal{L}_{\text{explain}}\left(\mathcal{M}_{ml}(\mathcal{P}(D_t), T) + \mathcal{V}_{ml}(\mathcal{P}(D_t), T) + \mathcal{S}_{ml}(\mathcal{P}(D_t), T)\right), \quad (10)$$

609 where $\mathcal{P}(\cdot)$ denotes the preprocessing operator, including data cleaning, anomaly
 610 handling, feature normalization, and validation procedures; $\mathcal{M}_{ml}$ represents the ML
 611 modeling component; $\mathcal{V}_{ml}$ denotes the visualization module; $\mathcal{S}_{ml}$ corresponds to
 612 statistical analysis; and $\mathcal{L}_{\text{explain}}$ denotes the LLM-driven interpretation layer that
 613 converts intermediate analytical outputs into human-readable scientific explanations.

Specifically, the framework receives four primary inputs: the dataset description D_{desc} , the task collection $T = \{t_1, t_2, \dots, t_n\}$, statistical metadata of the dataset S_{data} (e.g., variable types, dimensionality, sample size, and missing-value distributions), and a user-defined analytical objective G . For each task $t_i \in T$, it autonomously constructs a sequential analytical workflow encompassing visualization generation, statistical inference, ML modeling, and semantic interpretation. The complete procedure is orchestrated by the LLM without manual intervention, thereby substantially reducing human labor and improving analytical scalability.

For a given task t_i , the visualization stage first generates executable visualization code C_{vis} conditioned on $(t_i, D_{\text{desc}}, S_{\text{data}}, G)$. The generated code is subsequently executed to produce visualization results R_{vis} , including scatter plots, distribution maps, correlation matrices, feature projections, or temporal evolution diagrams, depending on task requirements. Formally,

$$C_{\text{vis}} = f_{\text{vis}}(t_i, D_{\text{desc}}, S_{\text{data}}, G) \text{ and } R_{\text{vis}} = \text{Execute}(C_{\text{vis}}), \quad (11)$$

where f_{vis} denotes the LLM-guided visualization code generator. Generated visual artifacts and intermediate outputs are automatically archived to the designated output repository P_{out} to ensure reproducibility and traceability.

Subsequently, the framework performs statistical analysis by generating analytical code C_{analysis} for hypothesis testing, correlation estimation, dimensionality reduction, clustering, or distributional analysis. The corresponding analytical results are represented as

$$C_{\text{analysis}} = f_{\text{analysis}}(t_i, D_{\text{desc}}, S_{\text{data}}, G) \text{ and } R_{\text{analysis}} = \text{Execute}(C_{\text{analysis}}). \quad (12)$$

In parallel with statistical inference, the modeling stage automatically determines suitable ML strategies according to task characteristics. Depending on the analytical objective, the framework may invoke supervised learning, unsupervised learning, ensemble methods, or regression-based predictive models. The generated modeling procedure can be expressed as

$$C_{\text{model}} = f_{\text{model}}(t_i, D_{\text{desc}}, S_{\text{data}}, G) \text{ and } R_{\text{model}} = \text{Execute}(C_{\text{model}}), \quad (13)$$

where f_{model} denotes the task-adaptive model generation function. Model outputs, including predictive metrics, latent representations, feature importance scores, and clustering structures, are stored for downstream interpretation and auditing.

To improve interpretability and readability, all successful intermediate outputs are further processed through an LLM-based semantic reasoning layer. Specifically, visualization outputs R_{vis} are transformed into descriptive interpretations A_{vis} ; statistical outputs R_{analysis} are converted into analytical summaries A_{analysis} ; and modeling outputs R_{model} are translated into model-level explanations A_{model} . This process can be represented as

$$A_x = \mathcal{L}_{\text{explain}}(R_x), \text{ where } x \in \{\text{vis}, \text{analysis}, \text{model}\}. \quad (14)$$

648 The interpretation layer is implemented using Gemini-Flash 2.0⁷ and DeepSeek-V3
 649 [45], enabling multimodal reasoning and automated scientific summarization across
 650 heterogeneous analytical outputs. Finally, the framework aggregates all intermediate
 651 analyses and semantic interpretations into a unified task-level report as

$$S_i = \Phi(A_{\text{vis}}, A_{\text{analysis}}, A_{\text{model}}), \quad (15)$$

652 where $\Phi(\cdot)$ denotes the report summarization operator. The generated report S_i ,
 653 together with all intermediate artifacts, is archived in P_{out} , while the task execution
 654 history is simultaneously updated within the historical workflow repository P_{his} .

655 4.2.3 Deep Learning Module

656 In astronomical data analysis, image data is an indispensable source of primary
 657 information. Its advantages lie in its visual representation and the abundance of
 658 multidimensional information it conveys, making it highly valuable for scientific
 659 interpretation. Researchers frequently leverage such image data to develop automated
 660 classification models to assist astronomers with large-scale identification tasks. How-
 661 ever, the development of these models has limitations; it typically requires meticulous
 662 design, including careful selection of model architecture, feature extraction meth-
 663 ods, training strategies, and performance evaluation metrics. Ensuring the model's
 664 robustness and generalizability across diverse observational conditions remains a
 665 key challenge, underscoring the complexity and importance of this research direc-
 666 tion in modern astronomy. In the design and implementation of this module, we
 667 mainly leverage the comprehensive algorithmic framework provided by MMPretrain
 668 [46], which significantly streamlines the configuration and deployment of state-of-
 669 the-art deep learning models. Given a collection of astronomical image data denoted
 670 as $D_i = \{I_1, I_2, \dots, I_k\}$ and task-specific network structure f_θ configured by $\mathcal{L}_{\text{config}}(T)$,
 671 the entire process can be simply represented as follows:

$$\mathcal{F}_{dl}(D_i, T) = f_\theta(D_i) + \mathcal{V}_{dl} \text{ and } f_\theta = \text{train}(D_i, Y). \quad (16)$$

672 Here, Y is the label associated with each image I_i , and $\mathcal{V}_{dl}$ is the visual report of the
 673 training process.

674 More specifically, the deep learning model construction process begins by ana-
 675 lyzing the input dataset description D_{desc} and the user-defined goal G . Based on
 676 these inputs, along with the available models in the repository M_{dir} , it selects a suit-
 677 able model architecture m_{model} and identifies the relevant data fields f_{field} through a
 678 `SelectModel` function. Subsequently, a configuration file m_{config} is generated via the
 679 `CreateConfig` procedure, encapsulating the selected model structure and training
 680 parameters. The configuration is then submitted to the training server for execution.
 681 If the training process completes successfully, the resulting trained model T and the
 682 configuration used are retrieved; otherwise, the process is flagged as failed, and fur-
 683 ther actions are halted. Once training is successful, both the model output and its
 684 configuration are saved to the designated output path P_{out} . Finally, it updates the

⁷<https://ai.google.dev/gemini-api>

historical execution path P_{his} with new information from the current task, thereby refining the goal G for future iterations. This pipeline streamlines the model development process by minimizing manual intervention and enabling the reuse of prior knowledge and model assets.

4.2.4 Time-Series Modeling Module

Similar to image data, time-series data is another essential modality in astronomy. It not only captures variations in light curves and spectral evolution but also enables investigations of explosion mechanisms and the inference of cosmological parameters. Therefore, the module is designed separately, although it also belongs to the category of tabular data. Given the time-series data $D_s \in \mathbb{R}^{n \times m \times t}$ (with t as the temporal dimension) and task T , a custom time-series model g_ϕ is configured via $\mathcal{L}_{\text{config}}(T)$:

$$\mathcal{F}_{ts}(D_s, T) = \begin{cases} g_\phi(D_s) + \mathcal{V}_s & \text{if } type == \text{other} \\ \mathcal{V}_{ts} + \mathcal{S}_{ts} & \text{otherwise} \end{cases} \quad \text{and } g_\phi = \text{train}(D_s, T). \quad (17)$$

Here, $\mathcal{V}_s$ represents the visualization report during training; $\mathcal{V}_{ts}$ and $\mathcal{S}_{ts}$ denote the visualization and statistical analysis modules, respectively.

For each task t_i in the task list T , the module distinguishes between general-purpose analyses and specialized time-series tasks. For the latter, the data is preprocessed via `PREPAREDATA`, and a configuration file is generated using `CREATECONFIG` based on the task type and data schema. The generated configuration is then submitted to the training server. If the execution is successful, the trained model and configuration details are collected and saved to the specified output path P_{out} . Additionally, the execution history is updated to support traceability and reproducibility. For tasks labeled as “others,” a default ML analysis is conducted.

4.2.5 Error Correction Mechanism

The error correction module in AutoAstro addresses unexpected events during analysis, focusing primarily on execution failures and data anomalies. To handle possible execution errors, the module employs an iterative error-correction mechanism that attempts to recover from failures up to a predefined maximum number of times. Specifically, given a code snippet, the module first writes it to a target file and executes it. If the execution returns an error, it captures the error message and leverages it to invoke a code correction module, which attempts to revise the original code based on the failure context (e.g., syntax or runtime errors tied to a specific column in the dataset). The corrected code is then re-executed in subsequent iterations. In cases where a system-level exception is raised prior to execution, AutoAstro similarly redirects the exception message into the correction workflow. The process repeats for a maximum of three attempts. If all attempts fail, it returns an error status; otherwise, it returns a success status. This mechanism ensures a degree of robustness and autonomy in the code generation and execution loop, enabling it to recover from a range of common runtime and logical errors without human intervention. When

faced with unclean data, this module is designed to perform essential data quality control operations on tabular datasets prior to downstream analysis. Specifically, it targets the detection and handling of column-level abnormalities, missing values, and duplicated records. The process begins with an optional user-invoked step for detecting abnormal columns, in which features exhibiting excessive proportions of invalid or placeholder values (e.g., “unknown”, “NA”, “-999”) are automatically removed if they exceed a predefined threshold. Following this, it systematically inspects missing values in both numeric and non-numeric columns. For numeric attributes, missing values are imputed with the median when a valid statistic can be computed. For non-numeric columns, users are prompted to either remove affected rows or defer manual correction. Finally, it identifies and removes duplicate records. In general, it offers an automated, interactive approach to data cleaning and serves as a foundational step in ensuring integrity for subsequent modeling or analysis.

Data availability

The datasets used in the experiments were obtained from public sources. Specifically, SpectrumCls was obtained from <https://doi.org/10.7717/peerj-cs.855>; Radio-GalaxyDataset was downloaded from <https://doi.org/10.5281/zenodo.7692494>; HTRU_Medlat was downloaded from <https://doi.org/10.1093/mnras/stu1188>; FAST_2018 was downloaded from <https://doi.org/10.1007/s11433-018-9388-3>; PLAsTiCC was obtained via Kaggle at <https://www.kaggle.com/competitions/PLAsTiCC-2018/data>; Kepler_Class was obtained from <https://doi.org/10.1088/1674-4527/20/10/167>; LAMOST DR10 was obtained from LAMOST at <https://lamost.org/dr10/>; Bright Star obtained via NASA at <https://heasarc.gsfc.nasa.gov/W3Browse/all/hipparcos.html>; celestial_object_classify was obtained via Kaggle at <https://www.kaggle.com/datasets/nitish200/astronomical-object-celestial-phenomena-catalog>; Gaia Astronomical Data were obtained via Kaggle at <https://www.kaggle.com/datasets/amit1235813/gaia-astronomical-data>; Kepler labeled time series were obtained via Kaggle at <https://www.kaggle.com/datasets/keplersmachines/kepler-labelled-time-series-data>; Simulated Hubble’s Law Data were obtained via Kaggle at <https://www.kaggle.com/datasets/austinhinkel/hubble-law-astronomy-lab>; Solar Flares from the RHESSI Mission were obtained via Kaggle at <https://www.kaggle.com/datasets/amit1235813/gaia-astronomical-data>; Star Types were obtained via Kaggle at <https://www.kaggle.com/datasets/deepu1109/star-dataset>; SDSS17 was obtained via Kaggle at <https://www.kaggle.com/datasets/fedesoriano/stellar-classification-dataset-sdss17>; Light curve of a variable star was obtained via LAMOST at <https://lamost.org/dr11/>; Tycho Astrometry Catalog was obtained via Kaggle at <https://www.kaggle.com/datasets/konivat/tycho-star-catalog>.

The NL2Astro dataset constructed in this study is available at the following URL: <https://doi.org/10.5281/zenodo.20799181>. All experimental results generated by the experiments in this study are available at the following URL: <https://doi.org/10.5281/zenodo.20806848>.

765 Code availability

766 The code for the experiments in this study is publicly available at the following URL:
767 <https://github.com/ACMISLab/AutoAstro>.

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774 References

- 775 [1] York, D.G., Adelman, J., Anderson, J.E. Jr., Anderson, S.F., Annis, J., Bahcall,
776 N.A., Bakken, J.A., Barkhouser, R., Bastian, S., Berman, E., *et al.*: The Sloan Digital
777 Sky Survey: Technical summary. *The Astronomical Journal* **120**(3), 1579 (2000)
- 778 [2] Zhao, G., Zhao, Y.-H., Chu, Y.-Q., Jing, Y.-P., Deng, L.-C.: LAMOST spectral
779 survey — an overview. *Research in Astronomy and Astrophysics* **12**(7), 723
780 (2012)
- 781 [3] Gaia Collaboration, *et al.*: Gaia data release 1. summary of the astromet-
782 ric, photometric, and survey properties. *Astronomy and Astrophysics* **595**(A2)
783 (2016)
- 784 [4] Ivezić, Ž., Kahn, S.M., Tyson, J.A., Abel, B., Acosta, E., Allsman, R., Alonso,
785 D., AlSayyad, Y., Anderson, S.F., Andrew, J., *et al.*: LSST: from science drivers
786 to reference design and anticipated data products. *The Astrophysical Journal*
787 **873**(2), 111 (2019)
- 788 [5] Jurić, M., Kantor, J., Lim, K.-T., Lupton, R.H., Dubois-Felsmann, G., Jenness,
789 T., Axelrod, T.S., Aleksić, J., Allsman, R.A., AlSayyad, Y., *et al.*: The LSST data
790 management system. In: *Astronomical Data Analysis Software and Systems*
791 **XXV**, pp. 279–288 (2016)
- 792 [6] Baby, B.E., Bhuvana, G., Radhika, D., Katoch, T., Mandal, S., Nandi, A.: Revealing
793 the nature of the transient source MAXI J0637-430 through spectro-temporal
794 analysis. *Monthly Notices of the Royal Astronomical Society* **508**(2), 2447–2457
795 (2021)
- 796 [7] Aigrain, S., Foreman-Mackey, D.: Gaussian process regression for astronomical
797 time series. *Annual Review of Astronomy and Astrophysics* **61**(1), 329–371 (2023)
- 798 [8] Stoppa, F., Bhattacharyya, S., Austri, R., Vreeswijk, P., Caron, S., Zaharijas, G.,
799 Bloemen, S., Principe, G., Malyshev, D., Vodeb, V., *et al.*: Autosourceid-classifier

- 800 – star-galaxy classification using a convolutional neural network with spatial
801 information. *Astronomy & Astrophysics* **680**, 109 (2023)
- 802 [9] Pieroni, M., Ricciardone, A., Barausse, E.: Detectability and parameter estimation
803 of stellar origin black hole binaries with next generation gravitational wave
804 detectors. *Scientific Reports* **12**, 17940 (2022)
- 805 [10] Parmiggiani, N., Bulgarelli, A., Ursi, A., Macaluso, A., Di Piano, A., Fioretti,
806 V., Aboudan, A., Baroncelli, L., Addis, A., Tavani, M., *et al.*: A deep-learning
807 anomaly-detection method to identify gamma-ray bursts in the ratemeters of the
808 agile anticoincidence system. *The Astrophysical Journal* **945**(2), 106 (2023)
- 809 [11] Neralwar, K.R., Colombo, D., Duarte-Cabral, A., Urquhart, J.S., Mattern, M.,
810 Wyrowski, F., Menten, K.M., Barnes, P., Sánchez-Monge, A., Beuther, H., *et al.*:
811 The SEDIGISM survey: Molecular cloud morphology-I. classification and star
812 formation. *Astronomy & Astrophysics* **663**, 56 (2022)
- 813 [12] Dutta, S., Paul, P.: Deep space insights: Machine learning revolutionizing
814 astrophysical discoveries. In: EPJ Web of Conferences, vol. 325, p. 01012 (2025)
- 815 [13] Meher, S.K., Panda, G.: Deep learning in astronomy: a tutorial perspective. *The*
816 *European Physical Journal Special Topics* **230**(10), 2285–2317 (2021)
- 817 [14] Barros, S., Demangeon, O., Diaz, R.F., Cabrera, J., Santos, N., Faria, J., Pereira, F.:
818 Improving transit characterisation with Gaussian process modelling of stellar
819 variability. *Astronomy & Astrophysics* **634**, 75 (2020)
- 820 [15] Sen, S., Agarwal, S., Chakraborty, P., Singh, K.P.: Astronomical big data process-
821 ing using machine learning: A comprehensive review. *Experimental Astronomy*
822 **53**(1), 1–43 (2022)
- 823 [16] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O.,
824 Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., *et al.*: Scikit-learn: Machine
825 learning in Python. *Journal of Machine Learning Research* **12**(85), 2825–2830
826 (2011)
- 827 [17] The Astropy Collaboration, Robitaille, T.P., Tollerud, E.J., Greenfield, P., Droett-
828 boom, M., Bray, E., Aldcroft, T., Davis, M., Ginsburg, A., Price-Whelan, A.M.,
829 *et al.*: Astropy: A community python package for astronomy. *Astronomy &*
830 *Astrophysics* **558**, 33 (2013)
- 831 [18] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T.,
832 Lin, Z., Gimelshein, N., Antiga, L., *et al.*: PyTorch: an imperative style, high-
833 performance deep learning library. *Advances in Neural Information Processing*
834 *Systems* **32**, 8026–8037 (2019)
- 835 [19] Riggi, S., Cecconello, T., Pilzer, A., Palazzo, S., Gupta, N., Hopkins, A., Trigilio,

- 836 C., Umana, G.: radio-llava: Advancing vision-language models for radio astro-
837 nomical source analysis. Publications of the Astronomical Society of Australia
838 **42**(e121) (2025)
- 839 [20] Wang, C., Hu, X., Zhang, Y., Chen, X., Du, P., Mao, Y., Wang, R., Li, Y., Wu,
840 Y., Yang, H., et al.: StarWhisper telescope: An AI framework for automating
841 end-to-end astronomical observations. arXiv **2412.06412** (2024)
- 842 [21] Kostunin, D., Sotnikov, V., Golovachev, S., Strube, A.: AI agents for ground-based
843 gamma astronomy. arXiv **2503.00821** (2025)
- 844 [22] OpenAI: GPT-4 technical report. arXiv **2303.08774** (2023)
- 845 [23] Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: ReAct: Syner-
846 gizing reasoning and acting in language models. In: The Eleventh International
847 Conference on Learning Representations (2023)
- 848 [24] Qiao, B., Li, L., Zhang, X., He, S., Kang, Y., Zhang, C., Yang, F., Dong, H., Zhang,
849 J., Wang, L., et al.: TaskWeaver: A code-first agent framework. arXiv **2311.17541**
850 (2023)
- 851 [25] OpenAI: Hello GPT-4o. <https://openai.com/index/hello-gpt-4o> (2024)
- 852 [26] Google: Gemini Deep Research – your personal research assistant. [https://](https://gemini.google/overview/deep-research/)
853 gemini.google/overview/deep-research/ (2025)
- 854 [27] Liu, X., Qin, B., Liang, D., Dong, G., Lai, H., Zhang, H., Zhao, H., Iong, I.L., Sun,
855 J., Wang, J., et al.: AutoGLM: Autonomous foundation agents for GUIs. arXiv
856 **2411.00820** (2024)
- 857 [28] VanderPlas, J., Connolly, A.J., Ivezić, Ž., Gray, A.: Introduction to astroML:
858 Machine learning for astrophysics. In: 2012 Conference on Intelligent Data
859 Understanding, pp. 47–54 (2012)
- 860 [29] Sahu, G., Puri, A., Rodriguez, J.A., Abaskohi, A., Chegini, M., Drouin, A.,
861 Taslakian, P., Zantedeschi, V., Lacoste, A., Vázquez, D., Chapados, N., Pal,
862 C., Rajeswar, S., Laradji, I.H.: InsightBench: Evaluating business analytics
863 agents through multi-step insight generation. In: The Thirteenth International
864 Conference on Learning Representations (2025)
- 865 [30] Guo, S., Deng, C., Wen, Y., Chen, H., Chang, Y., Wang, J.: DS-agent: Automated
866 data science by empowering large language models with case-based reasoning.
867 In: Proceedings of the 41st International Conference on Machine Learning, pp.
868 16813–16848 (2024)
- 869 [31] Haan, T., Ting, Y.-S., Ghosal, T., Nguyen, T.D., Accomazzi, A., Wells, A.,
870 Ramachandra, N., Pan, R., Sun, Z.: Achieving GPT-4o level performance in

- 871 astronomy with a specialized 8B-parameter large language model. Scientific
872 Reports 15, 13751 (2025)
- 873 [32] Iyer, K.G., Yunus, M., O'Neill, C., Ye, C., Hyk, A., McCormick, K., Ciucă, I.,
874 Wu, J.F., Accomazzi, A., Astarita, S., *et al.*: pathfinder: A semantic framework
875 for literature review and knowledge discovery in astronomy. The Astrophysical
876 Journal Supplement Series 275(2), 38 (2024)
- 877 [33] Evans, J., Sadruddin, S., D'Souza, J.: Astro-NER – astronomy named entity
878 recognition: Is GPT a good domain expert annotator? arXiv 2405.02602 (2024)
- 879 [34] Stoppa, F., Bulmus, T., Bloemen, S., Smartt, S.J., Groot, P.J., Vreeswijk, P.,
880 Smith, K.W.: Textual interpretation of transient image classifications from large
881 language models. Nature Astronomy 9, 1869–1878 (2025)
- 882 [35] Li, Y.-Y., Bai, Y., Wang, C., Qu, M., Lu, Z., Soria, R., Liu, J.: Deep learning and meth-
883 ods based on large language models applied to stellar light curve classification.
884 Intelligent Computing 4, 0110 (2025)
- 885 [36] Leung, H.W., Bovy, J.: Towards an astronomical foundation model for stars with
886 a transformer-based model. Monthly Notices of the Royal Astronomical Society
887 527(1), 1494–1520 (2024)
- 888 [37] Zhao, Y., Wang, J., Xiang, L., Zhang, X., Guo, Z., Turkay, C., Zhang, Y., Chen, S.:
889 LightVA: Lightweight visual analytics with LLM agent-based task planning and
890 execution. IEEE Transactions on Visualization and Computer Graphics 31(9),
891 6162–6177 (2024)
- 892 [38] Guo, J.-C., Yan, F.-B., Wan, G., Hu, X.-J., Wang, S.: A deep learning method for
893 the recognition of solar radio burst spectrum. PeerJ Compututer Science 8, 855
894 (2022)
- 895 [39] Rustige, L., Kummer, J., Griesse, F., Borrás, K., Brüggen, M., Connor, P.L., Gaede, F.,
896 Kasieczka, G., Knopp, T., Schleper, P.: Morphological classification of radio galax-
897 ies with wasserstein generative adversarial network-supported augmentation.
898 RAS Techniques and Instruments 2(1), 264–277 (2023)
- 899 [40] Wang, H., Zhu, W., Guo, P., Li, D., Feng, S., Yin, Q., Miao, C., Tao, Z., Pan, Z.,
900 Wang, P., *et al.*: Pulsar candidate selection using ensemble networks for FAST
901 drift-scan survey. Science China Physics, Mechanics & Astronomy 62, 1–10 (2019)
- 902 [41] Zeng, Q., Li, X., Lin, H.: Concat convolutional neural network for pulsar can-
903 didate selection. Monthly Notices of the Royal Astronomical Society 494(3),
904 3110–3119 (2020)
- 905 [42] Lu, H.-L., Li, Y.-Y., Li, Y.-B., Wang, C.-S., Luo, A.-L., Liang, J.-C., Li, S.: Spectra as
906 language: Large language models for scalable stellar parameter and abundance

- 907 inference. arXiv **2605.22162** (2026)
- 908 [43] Li, S., Yuan, G., Chen, J., Tan, C., Zhou, H.: Self-supervised learning for solar
909 radio spectrum classification. *Universe* **8**(12), 656 (2022)
- 910 [44] Fu, J.-N., Cat, P.D., Zong, W., Frasca, A., Gray, R.O., Ren, A.-B., Molenda-
911 Żakowicz, J., Corbally, C.J., Catanzaro, G., Shi, J.-R., *et al.*: Overview of the
912 LAMOST-Kepler project. *Research in Astronomy and Astrophysics* **20**(10), 167
913 (2020)
- 914 [45] DeepSeek-AI: DeepSeek-V3 technical report. arXiv **2412.19437** (2024)
- 915 [46] Chen, K., Wang, J., Pang, J., Cao, Y., Xiong, Y., Li, X., Sun, S., Feng, W., Liu, Z., Xu,
916 J., *et al.*: MMDetection: Open MMLab detection toolbox and benchmark. arXiv
917 **1906.07155** (2019)